



About

The **Centre for Learning Engineering and Sustainability Education (CLESE)** was established in **2024** with the vision of promoting innovative, inclusive, and sustainable approaches to education, with a special focus on **STEM (Science, Technology, Engineering and Mathematics) education**. The Centre serves as a platform for academic engagement, capacity building, knowledge sharing, and collaborative initiatives aimed at strengthening learning and teaching practices.

The Centre organizes **seminars, workshops, conferences, training programmes, and academic activities** in emerging areas of education, learning engineering, sustainability, and STEM. A major focus of the Centre is to provide **hands-on and experiential STEM learning opportunities** for school students, teachers, prospective teachers, scholars, and other stakeholders.

LEnSE has established collaborations with reputed institutions and organizations, including the **State Institute of Educational Technology (SIET), Government of Kerala, REFORM**, and the **Child Development Centre, Kazhakkootam**. These partnerships facilitate joint academic programmes, training initiatives, and educational outreach activities.

One of the key initiatives of the Centre is the establishment of a **dedicated STEM Learning Lab**, designed to provide practical, activity-based STEM training to school students. The Centre also plans to offer specialized STEM programmes for **prospective teachers**, enabling them to develop innovative classroom practices, strengthen pedagogical skills, and effectively integrate STEM approaches into teaching and learning.

The Centre follows an inclusive and socially responsible model in promoting STEM education. **Programme fees collected from its training activities are utilized to support STEM education for socially and economically disadvantaged students in rural schools across Kerala**, thereby creating opportunities for students who may have limited access to quality STEM learning resources.

LEnSE also seeks to expand its global academic engagement through an **international collaboration with the STEM Centre at Clarkson University, USA**. This collaboration is

intended to enhance the quality and scope of STEM training programmes, promote the exchange of knowledge and expertise, and introduce innovative STEM education practices.

The Centre conducts the **“Scholar Connect” series**, providing a platform for scholars to interact, share research ideas, discuss emerging academic developments, and explore collaborative research opportunities. In addition, CLESE has undertaken **collaborative projects with SIET, Government of Kerala, to provide STEM training to gifted students from 44 schools across Kerala**, supporting their intellectual curiosity, creativity, problem-solving abilities, and scientific thinking.

Through its diverse academic, training, research, and outreach activities, the **Centre for Learning Engineering and Sustainability Education** strives to build a strong ecosystem for innovative education, promote equitable access to STEM learning, empower teachers and learners, and contribute to the development of a scientifically literate, creative, and sustainable society.

Objectives:

1. To create educational environments using advanced computing capabilities to address career guidance at secondary level and professional employability at higher education level, to all irrespective of social economic cultural gender accessibility barriers.
2. To build new innovative digital tools that are helping catalyse improvements in students learning.
3. To initiate a comprehensive programme, based on state-of-the-art computer system of constructing mathematical models for use in education. To integrate psychological principles of human learning and AI with computational methods in order to widen the scope of educational sciences for aptitude development and career guidance at secondary level and employability skill development at higher education level.
4. To create and support opportunities for learning about sustainability through course work, research, co-curricular activities and civic engagement.
5. To improve opportunities for students to connect sustainability, global interdependence, equity and civic action in their academic studies and co-curricular activities and to provide research -informed professional development for teachers and school leaders are the major objectives of the Centre

A. Following are the programmes organized

1. State Institute of Educational Technology collaborated with Centre for Learning Engineering and Sustainability Education for the Fourth SIET International Conference on Educational Technology entitled “Educational Technology in STEAM Education”. The conference was conducted on 14th & 15th March 2025. Prof. Jan De Waters of Clarkson University was the chief guest and Hon’ble Minister for Education Sri. V.Sivankutty inaugurated the programme.
2. Invited Lecture by Dr.Briju Tankachan, Executive Director, EdTech Society & Visiting Professor, IIT Mumbai on the topic “ Prompt Engineering for Educators: ow to improve quality of Instruction” on 10th January 2025 at Department of Education, University of Kerala, Karyavattom Campus.
3. Conducted a two-month course on Selected Literary Terms as an extended learning programme aligned with the Ability Enhancement Course in Semester II, designed to support FYUGP learners under the University of Kerala. The course was developed by Ms. Greeshma Raveendran with guidance from Dr. Divya C. Senan and was offered through Centre for Learning Engineering & Sustainability Education. The course combined interactive video lessons, reading materials, quizzes, and assessments to promote self-directed, technology-enabled learning.
4. Conducted workshop on 'Shaping and nurturing future teachers of English at the secondary level; of education' in collaborating with ELTAI, the English Language Teachers' Association of India which is the oldest professional network of teachers of English in the country, and the Hornby Trust (UK) on 30 and 31st may 2025 at Centre for Under Graduate Studies, University of Kerala. The workshop aimed at bringing about transformative and impactful changes in the ways in which English is taught and learnt at the secondary level education.
5. The Centre for Learning Engineering and Sustainability Education at the University of Kerala collaborated with the Department of Education, University of Kerala, and the Indian Council for Social Science Research (ICSSR), New Delhi, to launch a state-wide initiative to conduct STEM (Science, Technology, Engineering, and Mathematics) camps in schools across Kerala for girls. This initiative is supported by funding from the ICSSR to promote STEM education and innovation among students. Through STEM-focused education, the STEM camp seeks to empower young women from economically and socially disadvantaged families in Kerala by offering skill development and evaluation programs that prepare them for careers in science. The ‘STEM 4 Girls’ Camp series were conducted in Govt. HSS, Puthoor, Kollam (District), Govt. HSS, Paruthippally, Thiruvananthapuram (District) and Govt HSS and VHSE, Nedumkandam, Idukki (District). The State level inauguration of the ‘STEM 4 Girls’ Camp series was done by Sri. KN Balagopal,

Hon'ble Minister for Finance, Govt. of Kerala. (State Level Programme implemented in one school from each district of Kerala).

6. The Centre for Learning Engineering and Sustainability Education at the University of Kerala collaborated with the State Institute of Educational Technology, functioning under the Department of General Education, Government of Kerala, successfully organized a three-day STEM (Science, Technology, Engineering, and Mathematics) camp from July 18 to 20, 2025 for 56 Gifted Students across Kerala. The camp was conducted at the Agriculture Cooperative Staff Training Institute, located in Manvila, Thiruvananthapuram. The State level inauguration of the 'STEM 4 Girls' Camp series was done by Sri. V Sivankutty, Hon'ble Minister for Education, Govt. of Kerala. (State Level Programme implemented among selected students from all educational districts of Kerala).
7. Organizing STEM Camps for Gifted Students across 41 educational districts of Kerala in collaboration with State Institute of Educational Technology. (State Level Programme implemented among selected students from all educational districts of Kerala).

Dr. Divya C Senan

Director

Centre for Learning Engineering and Sustainability Education

University of Kerala, Kariavattom Campus, Thiruvananthapuram-695581

Email: lenseedu24@gmail.com